Universal behaviour of the BCS energy gap

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Bardeen–Cooper–Schrieffer (BCS) 1957 theory of superconductivity

Described by **BCS gap equation** in $d = 1, 2, 3$ dimensions

$$\Delta(p) = -\frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} \hat{V}(p-q) \frac{\Delta(q)}{K_T^{\Delta}(q)} \, \mathrm{d}q, \qquad K_T^{\Delta}(p) = \frac{\sqrt{(p^2-\mu)^2 + |\Delta(p)|^2}}{\tanh\left(\frac{\sqrt{(p^2-\mu)^2+|\Delta(p)|^2}}{2T}\right)}.$$

Electron-lattice interaction leads to effective attractive electron-electron interaction V.

Material is superconducting if non-zero solution Δ exists. Equivalently (Hainzl–Hamza–Seiringer–Solovej 2008) if $\inf \operatorname{spec} K_T^0 + V < 0$.

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$\rightsquigarrow$ Define the **critical temperature** $T_c := \inf\{T \geq 0 : K_T^0 + V \geq 0\}$.

Phase transition at T_c: superconducting for $T < T_c$ and normal metal for $T \geq T_c$.

$E_\Delta(p) = \sqrt{(p^2-\mu)^2 + |\Delta(p)|^2}$ dispersion relation for associated quasi-particles.

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T_c and Ξ are material-dependent (μ, V-dependent). Both measure "stability" of superconducting phase.

Question

- Formula for T_c and Ξ in various limits?
- Relation between T_c and Ξ ?

Zero temperature

Physics prediction for T_c small:

$$\frac{\Xi(T=0)}{T_c} \approx \pi e^{-\gamma} \approx 1.76$$

Well-known in physics literature.

Valid for s-wave superconductivity (Δ radial),
(Ex: V radial with $\hat{V} \leq 0$)

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- Weak coupling λV with $\lambda \to 0$,
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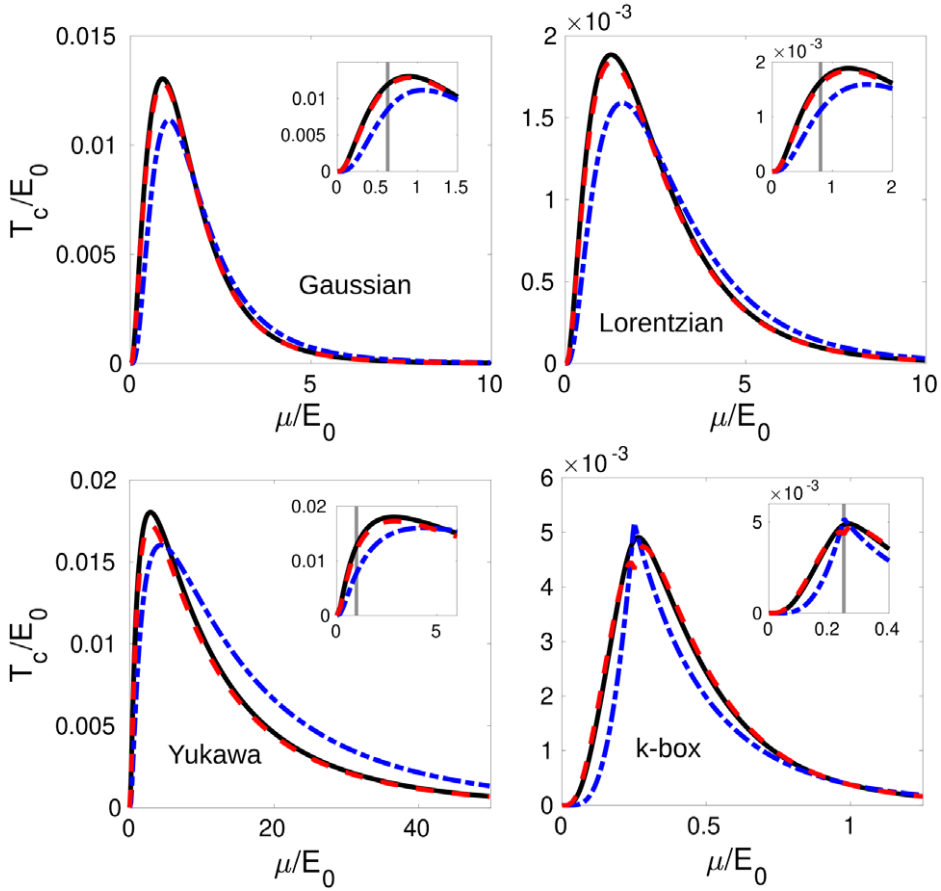

Figure: From "E. Langmann, C. Triola, and A. V. Balatsky. Phys. Rev. Lett. **122**.15 (2019)"

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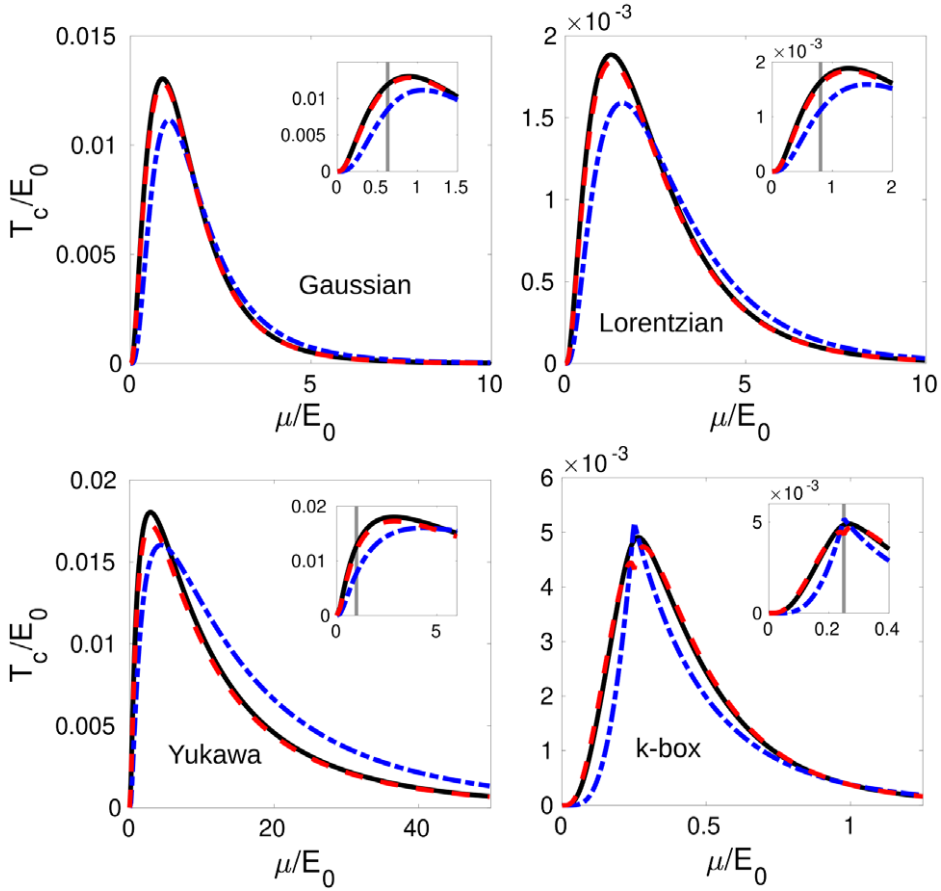
Can we find T_c and $\Xi(T=0)$ in these limits?



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- *in dimensions $d = 1, 2, 3$ for interaction λV in limit $\lambda \to 0$ (weak coupling):*

$$T_c = \mu \left(2c_d \frac{e^\gamma}{\pi} + o(1) \right) \exp\left(\frac{\pi}{2\sqrt{\mu}^{d-2} b_\mu^{(d)}(\lambda)} \right), \quad \Xi = \mu \left(2c_d + o(1) \right) \exp\left(\frac{\pi}{2\sqrt{\mu}^{d-2} b_\mu^{(d)}(\lambda)} \right),$$

[Frank–Hainzl–Naboko–Seiringer 2007; Hainzl–Seiringer 2008b; Henheik–L.–Roos 2024],

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In all limits: $$\frac{\Xi(T=0)}{T_c} = \pi e^{-\gamma}(1 + o(1))$$

"General" result for $d = 3$ and s-wave superconductivity [Langmann–Triola–Balatsky 2019; Langmann–Triola 2023].

Non-zero temperature

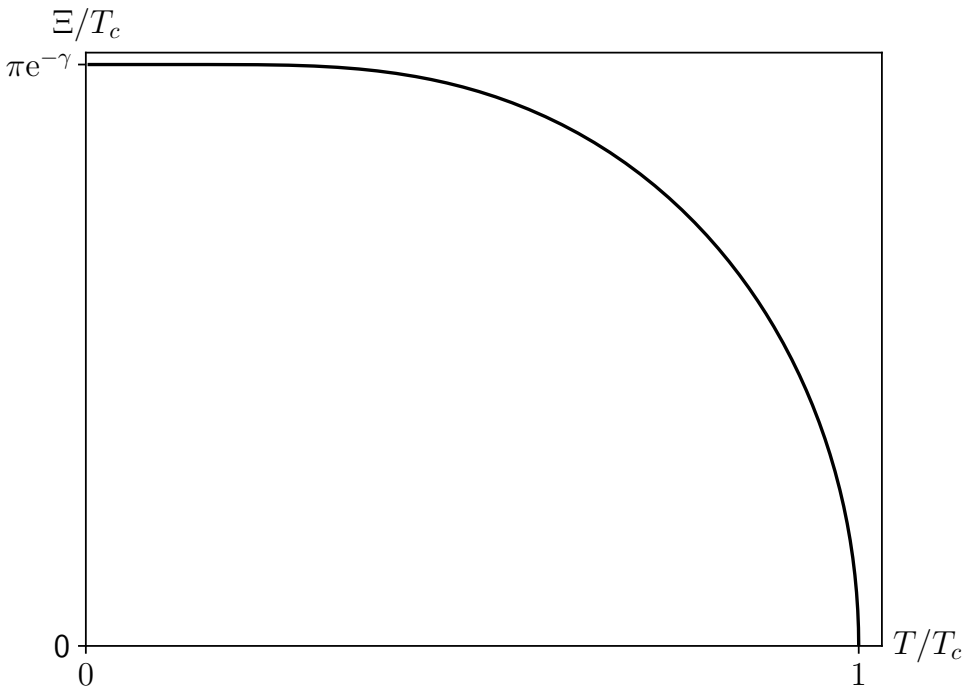
Physics prediction for T_c small (for s-wave superconductivity):

$$\frac{\Xi(T)}{T_c} \approx \mathrm{f}_{\mathrm{BCS}}\left(\sqrt{1-\frac{T}{T_c}}\right)$$

for **universal** function $\mathrm{f}_{\mathrm{BCS}}$.
For $T=0$ and $T \to T_c$:

$$\frac{\Xi(T=0)}{T_c} \approx \pi e^{-\gamma} \approx 1.76,$$

$$\frac{\Xi(T)}{T_c} \approx \sqrt{\frac{8\pi^2}{7\zeta(3)}}\sqrt{1-\frac{T}{T_c}} \approx 3.06\sqrt{1-\frac{T}{T_c}}$$



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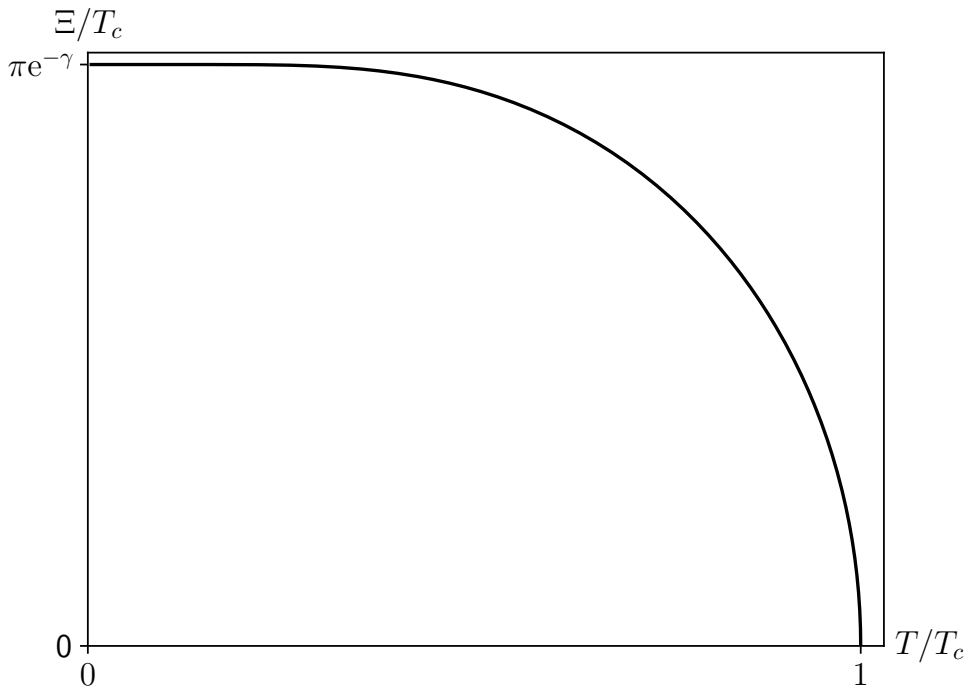

Theorem (Langmann–Triola 2023, Henheik–L. 2025)

Picture holds

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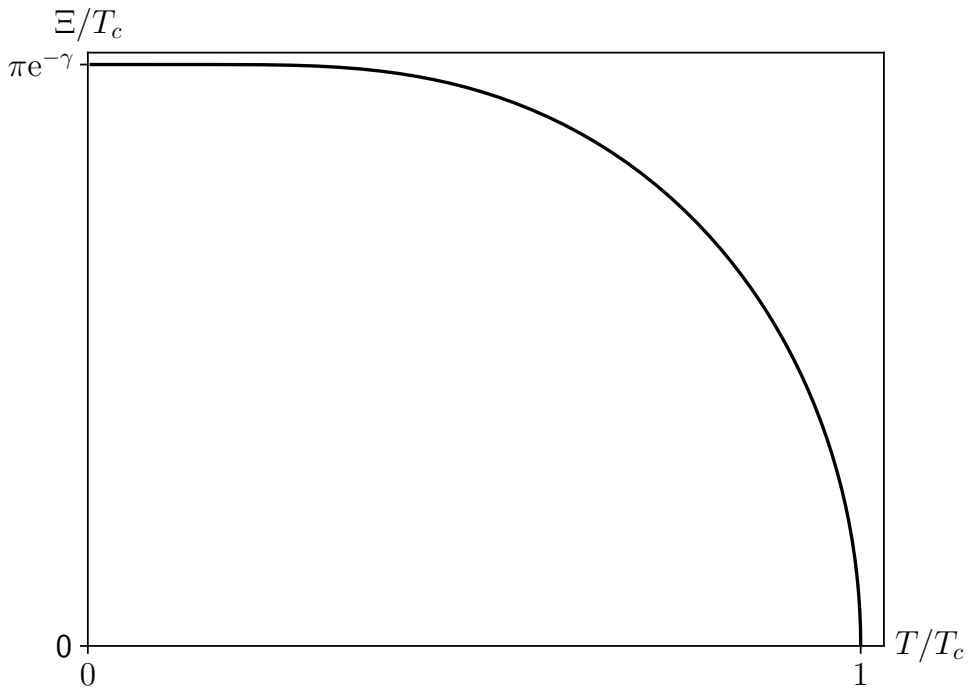

Theorem (Langmann–Triola 2023, Henheik–L. 2025)

Picture holds for interaction λV radial in limit $\lambda \to 0$

- *in dimensions $d = 1, 2, 3$ for V with $\hat{V} \leq 0$*
  (s-wave superconductivity, weak coupling limit),
- *in dimension $d = 2$ for V with* $\dim\ker(K_{T_c} + \lambda V) \leq 2$
  (pure angular momentum, weak coupling limit),